ETALE LOCAL FLATNESS AFTER ADMISSIBLE BLOWUPS GLOBALISES TO ONE ADMISSIBLE BLOWUP MAKING THE STRICT TRANSFORM FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0814 (sources/stacks-flat/pinned-original-source.tex line 8904).

Allowed standard prerequisites (editorial).

- Strict transforms and flatness under blowups and blowup composition. Divisors, Lemmas labelled divisors-lemma-strict-transform-flat and divisors-lemma-strict-transform-composition-blowups, cited at proof lines 8958-8961; Divisors, Lemma labelled divisors-lemma-flat-base-change-blowing-up, cited at proof lines 8942-8944. Tracks the strict transform of the module through the base change and the additional blowup.
- Comparison of ideals in two blowups. Divisors, Lemma labelled divisors-lemma-blowing-up-two-ideals, cited at proof lines 8946-8948 and 9002-9004. Produces the commutative square of blowups and shows the base change of the final blowup factors through the prescribed one.
- Successive blowups and reduction to the affine divisor case. Same chapter, Remark labelled remark-successive-blowups, cited at proof lines 8974-8978; Divisors, Lemma labelled divisors-lemma-characterize-effective-Cartier-divisor, cited at proof lines 8978-8980. Reduces to S = Spec(A) with S minus U the zero locus of a nonzerodivisor.
- Splitting an ideal after an etale base change. Same chapter, Lemma labelled lemma-push-ideal, cited at proof lines 8984-8986. Produces the finite type ideals I_i on S whose pullbacks split off the given ideals J_i.
- Flatness of a module and its pullback along an etale cover. Same chapter, Lemma labelled lemma-etale-flat-up-down, cited at proof lines 9008-9010; Morphisms, Lemma labelled morphisms-lemma-flat-permanence, cited at

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proof lines 9010-9012; Morphisms, Lemma labelled morphisms-lemma-base-change-module-flat, cited at proof lines 8968-8970. Descends flatness from the etale cover S_i back to S.

Source reference. Original lemma 0805: sources/stacks-chow/tags.txt, source lines 10170–10170. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8940 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080A: sources/stacks-chow/tags.txt, source lines 10175–10175. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8944 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080F: sources/stacks-chow/tags.txt, source lines 10180–10180. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8953 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080I: sources/stacks-chow/tags.txt, source lines 10183–10183. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8954 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01U8: sources/stacks-chow/tags.txt, source lines 2401–2401. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8970 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 080Y: sources/stacks-flat/pinned-original-source.tex, source lines 8611–8611. Same-chapter pointer into the author environment remark at source line 8611; invoked at proof line 8972 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01WS: sources/stacks-chow/tags.txt, source lines 2493–2493. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8979 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0813: sources/stacks-flat/pinned-original-source.tex, source lines 8876–8876. Same-chapter pointer into the author environment lemma at source line 8876; invoked at proof line 8984 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 05B9: sources/stacks-flat/pinned-original-source.tex, source lines 124–124. Same-chapter pointer into the author environment lemma at source line 124; invoked at proof line 9011 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02JZ: sources/stacks-chow/tags.txt, source lines 3354–3354. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9012 of the counted proof. It is

retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let $X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent module on X. Let $U \subset S$ be a quasi-compact open. Assume there exist finitely many commutative diagrams*

$$\begin{array}{ccccc} & & X_i & \xrightarrow[j_i]{} & X \\ & & \downarrow & & \downarrow \\ S_i^* & \longrightarrow & S_i & \xrightarrow{e_i} & S \end{array}$$

where

1. *$e_i : S_i \to S$ are quasi-compact étale morphisms and $S = \bigcup e_i(S_i)$,*
2. *$j_i : X_i \to X$ are étale morphisms and $X = \bigcup j_i(X_i)$,*
3. *$S_i^* \to S_i$ is an $e_i^{-1}(U)$-admissible blowup such that the strict transform $\mathcal{F}_i^*$ of $j_i^*\mathcal{F}$ is flat over S_i^*.*

Then there exists a U-admissible blowup $S' \to S$ such that the strict transform of $\mathcal{F}$ is flat over S'.

Proof. We claim that the hypotheses of the lemma are preserved under U-admissible blowups. Namely, suppose $b : S' \to S$ is a U-admissible blowup in the quasi-coherent sheaf of ideals $\mathcal{I}$. Moreover, let $S_i^* \to S_i$ be the blowup in the quasi-coherent sheaf of ideals $\mathcal{J}_i$. Then the collection of morphisms $e_i' : S_i' = S_i \times_S S' \to S'$ and $j_i' : X_i' = X_i \times_S S' \to X \times_S S'$ satisfy conditions (1), (2), (3) for the strict transform $\mathcal{F}'$ of $\mathcal{F}$ relative to the blowup $S' \to S$. First, observe that S_i' is the blowup of S_i in the pullback of $\mathcal{I}$, see Divisors, Lemma 0805. Second, consider the blowup $S_i'^* \to S_i'$ of S_i' in the pullback of the ideal $\mathcal{J}_i$. By Divisors, Lemma 080A we get a commutative diagram

$$\begin{array}{ccc} S_i'^* & \longrightarrow & S_i' \\ \downarrow & \searrow & \downarrow \\ S_i^* & \longrightarrow & S_i \end{array}$$

and all the morphisms in the diagram above are blowups. Hence by Divisors, Lemmas 080F and 080I we see

$$\begin{aligned} & \text{the strict transform of } (j_i')^*\mathcal{F}' \text{ under } S_i'^* \to S_i' \\ = \ & \text{the strict transform of } j_i^*\mathcal{F} \text{ under } S_i'^* \to S_i \\ = \ & \text{the strict transform of } \mathcal{F}_i' \text{ under } S_i'^* \to S_i' \\ = \ & \text{the pullback of } \mathcal{F}_i^* \text{ via } X_i \times_{S_i} S_i'^* \to X_i \end{aligned}$$

which is therefore flat over $S_i'^*$ (Morphisms, Lemma 01U8). Having said this, we see that all observations of Remark 080Y apply to the problem of finding a U-admissible blowup such that the strict transform of $\mathcal{F}$ becomes flat over the base under assumptions as in the lemma. In particular, we may assume that $S \setminus U$ is the support of an effective Cartier divisor $D \subset S$. Another application of Remark 080Y combined with Divisors, Lemma 01WS shows we may assume that $S = \mathrm{Spec}(A)$ and $D = \mathrm{Spec}(A/(a))$ for some nonzerodivisor $a \in A$.

Pick an i and $s \in S_i$. Lemma 0813 implies we can find an open neighbourhood $s \in W_i \subset S_i$ and a finite type quasi-coherent ideal $\mathcal{I} \subset \mathcal{O}_S$ such that $\mathcal{I} \cdot \mathcal{O}_{W_i} = \mathcal{J}_i\mathcal{J}_i'$ for some finite type quasi-coherent ideal $\mathcal{J}_i' \subset \mathcal{O}_{W_i}$ and such that $V(\mathcal{I}) \subset V(a) = S \setminus U$. Since S_i is quasi-compact we can replace S_i by a finite collection $W_1, \ldots, W_n$ of

these opens and assume that for each i there exists a quasi-coherent sheaf of ideals $\mathcal{I}_i \subset \mathcal{O}_S$ such that $\mathcal{I}_i \cdot \mathcal{O}_{S_i} = \mathcal{J}_i\mathcal{J}_i'$ for some finite type quasi-coherent ideal $\mathcal{J}_i' \subset \mathcal{O}_{S_i}$. As in the discussion of the first paragraph of the proof, consider the blowup S' of S in the product $\mathcal{I}_1 \ldots \mathcal{I}_n$ (this blowup is U-admissible by construction). The base change of $S' \to S$ to S_i is the blowup in

$$\mathcal{J}_i \cdot \mathcal{J}_i'\mathcal{I}_1 \ldots \hat{\mathcal{I}}_i \ldots \mathcal{I}_n$$

which factors through the given blowup $S_i^* \to S_i$ (Divisors, Lemma 080A). In the notation of the diagram above this means that $S_i'^* = S_i'$. Hence after replacing S by S' we arrive in the situation that $j_i^*\mathcal{F}$ is flat over S_i. Hence $j_i^*\mathcal{F}$ is flat over S, see Lemma 05B9. By Morphisms, Lemma 02JZ we see that $\mathcal{F}$ is flat over S. $\square$